Reducing 3SAT to Zero-Finding for Constant-Jacobian Polynomial Maps

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Abstract

We give an explicit polynomial-time reduction from 3SAT to the problem of deciding whether a polynomial map

$$K : \mathbb{R}^N \longrightarrow \mathbb{R}^N$$

has a real zero, under the promise that its Jacobian determinant is a fixed nonzero constant.

The reduction uses a single fixed polynomial map

$$H : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

with three properties:

1. $\det JH$ is a nonzero constant;
2. the preimage $H^{-1}(0)$ consists of three explicitly known points encoding the bits 0 and 1;
3. one explicitly known target Δ is omitted from the image of H.

Copies of this fixed map encode variables and clauses. The Jacobian of the resulting map is block lower triangular, so its determinant is immediately seen to be a nonzero constant.

1 Statement of the reduction

Theorem 1. *There is a polynomial-time algorithm which, given a 3CNF formula ϕ with n variables and m clauses, outputs an integer-coefficient polynomial map*

$$K_\phi : \mathbb{R}^{3n+3m} \longrightarrow \mathbb{R}^{3n+3m}$$

such that:

1.
$$\det JK_\phi \equiv (-3456)^{n+m};$$

2.
$$\deg K_\phi \leq 7;$$

3.
$$\phi \in \mathrm{SAT} \quad \Longleftrightarrow \quad \exists u \in \mathbb{R}^{3n+3m} \text{ such that } K_\phi(u) = 0.$$

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Consequently, since 3SAT *is NP-hard [3], deciding real zero existence for integer-coefficient polynomial maps with constant nonzero Jacobian determinant is NP-hard.*

The constant-Jacobian problem goes back to Keller's original paper [4]. Our proof is based on one fixed three-dimensional gadget. Its unscaled form is the explicit map announced by Alpöge [1]; the affine rescaling below is convenient for the reduction.

2 The fixed polynomial gadget

Define

$$H = (H_1, H_2, H_3) : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

by

$$H_1(x, y, z) = 12\left((1 + xy)^3 z + y^2(1 + xy)(4 + 3xy)\right) + 3, \tag{1}$$
$$H_2(x, y, z) = 12\left(y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy)\right), \tag{2}$$
$$H_3(x, y, z) = 12\left(2x - 3x^2 y - x^3 z\right). \tag{3}$$

Also define the three points

$$r_0 = \left(0, 0, -\frac{1}{4}\right), \qquad r_+ = \left(1, -\frac{3}{2}, \frac{13}{2}\right), \qquad r_- = \left(-1, \frac{3}{2}, \frac{13}{2}\right),$$

and the target

$$\Delta = (7, 24, 8).$$

For a target $q \in \mathbb{R}^3$, its preimage under H[1] is

$$H^{-1}(q) = \{w \in \mathbb{R}^3 : H(w) = q\}.$$

Lemma 2 (Fixed gadget)**.** *The map H satisfies:*

1.
$$\det JH \equiv -3456;$$

2.
$$H^{-1}(0) = r_0, r_+, r_-;$$

3.
$$H^{-1}(\Delta) = \varnothing.$$

Proof. The Jacobian identity is a direct polynomial calculation:

$$\det JH = 12^3(-2) = -3456.$$

For completeness, the two preimage claims have short Gröbner-basis certificates. We use the standard connection between Gröbner bases, elimination, and polynomial solution sets; see, for example, [2, Chs. 2–4].

With respect to lexicographic order

$$z \succ y \succ x,$$

[1]In standard terminology, this preimage is called the *fiber* of H over q.

a Gröbner basis for the ideal defining the preimage of zero,

$$\langle H_1, H_2, H_3 \rangle \subseteq \mathbb{Q}[x, y, z]$$

is

$$\left\{ z - \frac{27}{4}x^2 + \frac{1}{4},\ y + \frac{3}{2}x,\ x^3 - x \right\}.$$

Hence every zero satisfies

$$x^3 - x = x(x-1)(x+1) = 0,$$

so

$$x \in \{-1, 0, 1\}.$$

The other two equations then uniquely determine y and z:

$$y = -\frac{3}{2}x, \qquad z = \frac{27}{4}x^2 - \frac{1}{4}.$$

The resulting three points are exactly

$$r_0, \qquad r_+, \qquad r_-.$$

For the omitted target, a Gröbner basis for

$$\langle H_1 - 7,\ H_2 - 24,\ H_3 - 8 \rangle$$

is

$$\{1\}.$$

Therefore this system has no solution even over $\mathbb{C}$, and in particular has no real solution. □

3 The two jobs of the fixed gadget

The reduction uses H as a small machine with two independent jobs.

First, the preimage of zero under H stores one Boolean bit. Define

$$\beta(x, y, z) = x^2.$$

The three possible zeroes of H decode as

$$\beta(r_0) = 0, \qquad \beta(r_+) = \beta(r_-) = 1.$$

Consequently,

$$H(w) = 0 \quad \Longrightarrow \quad \beta(w) \in \{0, 1\}.$$

Both bit values can be encoded: use r_0 for 0, and use either r_+ or r_- for 1.

Second, H can hit the target 0 but cannot hit the target Δ. This lets us turn a Boolean value into a question about whether an equation has a solution.

3.1 The target selector

For a scalar s, define the map $T_s : \mathbb{R}^3 \to \mathbb{R}^3$ by

$$T_s(v) = H(v) - (1 - s)\Delta.$$

The key observation is simply

$$T_s(v) = 0 \quad \Longleftrightarrow \quad H(v) = (1 - s)\Delta.$$

Thus s selects the target that H is asked to hit. For Boolean s there are only two cases:

s	target required from H	can H hit it?
1	0	yes
0	Δ	no.

Lemma 3 (Boolean target selector)**.** *For $s \in \{0, 1\}$,*

$$T_s^{-1}(0) \neq \varnothing \quad \Longleftrightarrow \quad s = 1.$$

Moreover,

$$\det D_v T_s(v) \equiv -3456.$$

Proof. If $s = 1$, then $T_1 = H$, so for example $T_1(r_0) = 0$. If $s = 0$, then $T_0(v) = 0$ would imply $H(v) = \Delta$, which is impossible by the fixed-gadget lemma. Finally, $(1 - s)\Delta$ is constant with respect to v, so

$$D_v T_s(v) = JH(v).$$

The determinant claim follows from $\det JH \equiv -3456$. □

Nothing is claimed about non-Boolean values of s. In the reduction below, the variable equations ensure that every value fed into a target selector is either 0 or 1.

4 From a formula to a polynomial map

Let $\phi(X_1, \ldots, X_n)$ be a 3CNF formula with clauses $C_1, \ldots, C_m$. The construction has three steps:

1. force each variable block to encode a bit;
2. compute the Boolean value of every clause as a polynomial;
3. use a target selector to require every clause value to be 1.

4.1 Variables

For each variable X_i, introduce a block

$$w_i = (a_i, b_i, c_i) \in \mathbb{R}^3$$

and include $H(w_i)$ in the output. At a zero of the final map this forces $w_i \in \{r_0, r_+, r_-\}$, so

$$X_i = \beta(w_i)$$

defines a Boolean assignment. Conversely, an assignment can be encoded by choosing $w_i = r_0$ when $X_i = 0$ and $w_i = r_+$ when $X_i = 1$.

4.2 Literals and clauses

For a literal ℓ, define its polynomial value by

$$\operatorname{val}(\ell) = \begin{cases} \beta(w_i), & \ell = X_i, \\ 1 - \beta(w_i), & \ell = \neg X_i. \end{cases}$$

Whenever $H(w_i) = 0$, each such literal value belongs to $\{0, 1\}$ and agrees with the usual Boolean evaluation under the assignment

$$X_i = \beta(w_i).$$

Let

$$C_j = \ell_{j,1} \vee \ell_{j,2} \vee \ell_{j,3}$$

be the jth clause. Define its satisfaction polynomial

$$s_j(w_1, \ldots, w_n) = 1 - \prod_{k=1}^{3} \left(1 - \operatorname{val}(\ell_{j,k})\right).$$

On Boolean inputs,

$$s_j \in \{0, 1\},$$

and

$$s_j = 1 \quad \Longleftrightarrow \quad C_j \text{ is satisfied.}$$

For each clause, introduce a fresh witness block

$$v_j \in \mathbb{R}^3$$

and include the output

$$T_{s_j}(v_j) = H(v_j) - (1 - s_j)\Delta.$$

Because $s_j \in \{0, 1\}$ at every solution of the variable equations, the target-selector lemma implies

$$\exists v_j : \; T_{s_j}(v_j) = 0 \quad \Longleftrightarrow \quad s_j = 1.$$

In other words, v_j is a witness that exists exactly when clause C_j is satisfied.

4.3 The final map

Let ϕ have n variables and m clauses. Define

$$K_\phi : \mathbb{R}^{3n+3m} \longrightarrow \mathbb{R}^{3n+3m}$$

by

$$K_\phi\left(w_1, \ldots, w_n, v_1, \ldots, v_m\right) = \begin{pmatrix} H(w_1) \\ \vdots \\ H(w_n) \\ T_{s_1}(v_1) \\ \vdots \\ T_{s_m}(v_m) \end{pmatrix}.$$

Equivalently,

$$K_\phi = \begin{pmatrix} H(w_1) \\ \vdots \\ H(w_n) \\ H(v_1) - (1 - s_1(w))\Delta \\ \vdots \\ H(v_m) - (1 - s_m(w))\Delta \end{pmatrix}.$$

5 Why the reduction works

The construction was designed so that its zeroes and satisfying assignments are the same information in two different forms.

Lemma 4 (Correctness)**.** *The map K_ϕ has a real zero if and only if ϕ is satisfiable.*

Proof. Suppose first that $K_\phi(w, v) = 0$. Every variable block then satisfies $H(w_i) = 0$, so the values $X_i = \beta(w_i)$ form a Boolean assignment. Every clause block satisfies

$$T_{s_j}(v_j) = 0.$$

Since s_j is Boolean, the target-selector lemma forces $s_j = 1$. Hence every clause is true under the decoded assignment, and ϕ is satisfiable.

Conversely, suppose that ϕ has a satisfying assignment σ. Encode it by choosing

$$w_i = \begin{cases} r_0, & \sigma(X_i) = 0, \\ r_+, & \sigma(X_i) = 1. \end{cases}$$

Then $H(w_i) = 0$ and $\beta(w_i) = \sigma(X_i)$. Since σ satisfies every clause, every $s_j = 1$. Choose $v_j = r_0$ for all j; then

$$T_{s_j}(v_j) = T_1(r_0) = H(r_0) = 0.$$

Thus every output coordinate of K_ϕ vanishes. We have proved

$$K_\phi^{-1}(0) \neq \varnothing \quad \Longleftrightarrow \quad \phi \in \mathrm{SAT}.$$

□

6 Completing the proof of the reduction

It remains to check that the construction has the promised Jacobian, degree, coefficient, and complexity bounds.

Lemma 5 (Constant Jacobian)**.** *The Jacobian determinant of K_ϕ is the nonzero constant*

$$\det JK_\phi \equiv (-3456)^{n+m}.$$

Proof. Order the input blocks as

$$w_1, \ldots, w_n, v_1, \ldots, v_m.$$

The intuition is that the clause gadgets may look backward at the encoded variables, but no variable gadget looks forward at the clause witnesses. Algebraically, the variable outputs depend only on the w-blocks. The jth clause output may depend on all w-blocks through s_j, but among the witness blocks it depends only on v_j.

Therefore the Jacobian has block lower-triangular form

$$JK_\phi = \begin{pmatrix} D_{\text{var}} & 0 \\ B & D_{\text{clause}} \end{pmatrix},$$

where

$$D_{\text{var}} = \operatorname{diag}\left(JH(w_1), \ldots, JH(w_n)\right)$$

and

$$D_{\text{clause}} = \operatorname{diag}\left(JH(v_1), \ldots, JH(v_m)\right).$$

The lower-left block B contains derivatives of the s_j with respect to the variable blocks. Its exact form is irrelevant to the determinant. The diagonal clause blocks are still $JH(v_j)$ because the target shift $(1 - s_j)\Delta$ does not depend on v_j.

Since the determinant of a block triangular matrix is the product of the determinants of its diagonal blocks,

$$\begin{aligned} \det JK_\phi &= \det D_{\text{var}} \det D_{\text{clause}} && (4) \\ &= \prod_{i=1}^{n} \det JH(w_i) \prod_{j=1}^{m} \det JH(v_j) && (5) \\ &= (-3456)^n(-3456)^m && (6) \\ &= (-3456)^{n+m}. && (7) \end{aligned}$$

Thus the Jacobian determinant is a nonzero constant polynomial. □

Lemma 6 (Degree, coefficients, and size)**.** *The map K_ϕ has integer coefficients, degree at most 7, and a description that can be constructed in time polynomial in the size of ϕ.*

Proof. The map H has degree 7, while β has degree 2. Each s_j is a product of at most three degree-2 literal expressions, so

$$\deg s_j \leq 6.$$

Each clause output has the form

$$H(v_j) - (1 - s_j)\Delta.$$

Therefore

$$\deg K_\phi \leq \max\{7, 6\} = 7.$$

All coefficients are integers:

- H has integer coefficients;
- $\Delta = (7, 24, 8)$ is integral;
- every s_j has integer coefficients.

There are exactly

$$3n + 3m$$

input and output coordinates. Since each variable and clause contributes only a constant-size polynomial block, the description of K_ϕ has size polynomial in the description length of ϕ. □

Proof of Theorem 1. The correctness lemma gives

$$K_\phi^{-1}(0) \neq \varnothing \quad \Longleftrightarrow \quad \phi \in \mathrm{SAT}.$$

The two preceding lemmas give the claimed Jacobian, degree, coefficient, and polynomial-time construction bounds. Since 3SAT is NP-hard [3], the zero-existence problem in the theorem is NP-hard. □

A Exact symbolic verification

The only special algebraic facts are:

$$\det JH = -3456, \qquad H^{-1}(0) = r_0, r_+, r_-, \qquad H^{-1}(\Delta) = \varnothing.$$

They can be verified exactly over $\mathbb{Q}$ with the following script.

```
import sympy as sp

x, y, z = sp.symbols("x␣y␣z")
u = 1 + x*y

P = u**3*z + y**2*u*(4 + 3*x*y)
Q = y + 3*x*u**2*z + 3*x*y**2*(4 + 3*x*y)
R = 2*x - 3*x**2*y - x**3*z

F = sp.Matrix([P, Q, R])

# The unscaled map has constant Jacobian -2.

assert sp.factor(F.jacobian((x, y, z)).det()) == -2

# H = 12(F - (-1/4,0,0)).

H = sp.Matrix([
12*P + 3,
12*Q,
12*R,
])

assert sp.factor(H.jacobian((x, y, z)).det()) == -3456

# Preimage of zero under H:

# equivalent to F = (-1/4,0,0).

G_zero = sp.groebner(
[
P + sp.Rational(1, 4),
Q,
R,
],
z, y, x,
order="lex",
```

```
)

print(G_zero)

# z - 27*x**2/4 + 1/4

# y + 3*x/2

# x**3 - x

# Preimage of (7,24,8) under H:

# equivalent to F = (1/3,2,2/3).

G_omitted = sp.groebner(
[
P - sp.Rational(1, 3),
Q - 2,
R - sp.Rational(2, 3),
],
z, y, x,
order="lex",
)

print(G_omitted)

# 1
```

References

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